

EXPOSED LOFT FLOOR · JOIST / W-BEAM CONNECTION STUDY

Garage study · W12×16 concept · 2×8 joists at 16" o.c. · preliminary architectural details

JH-001

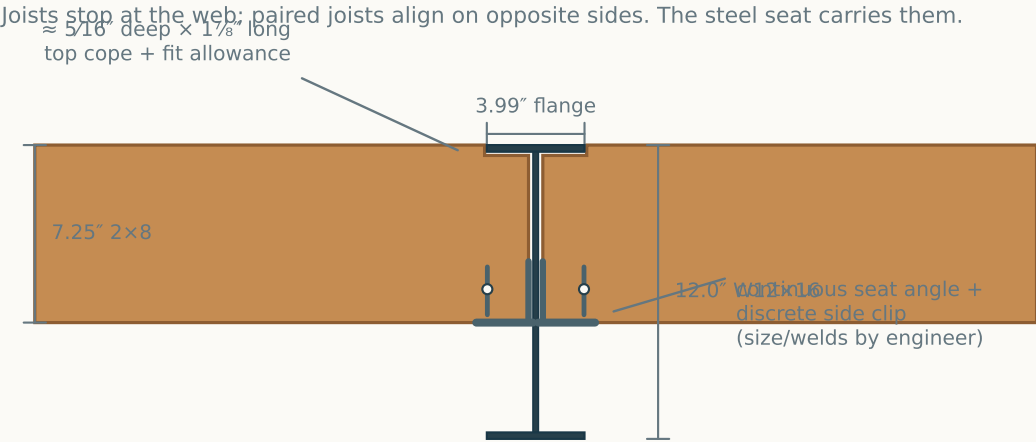
21 SEPTEMBER 2026

DESIGN INTENT

quiet painted wood cells, dark steel lines, flush top plane, wiring and low-profile lights hidden above

NOT FOR CONSTRUCTION

A FLUSH TUCKED JOIST — CLEANEST CUSTOM DETAIL



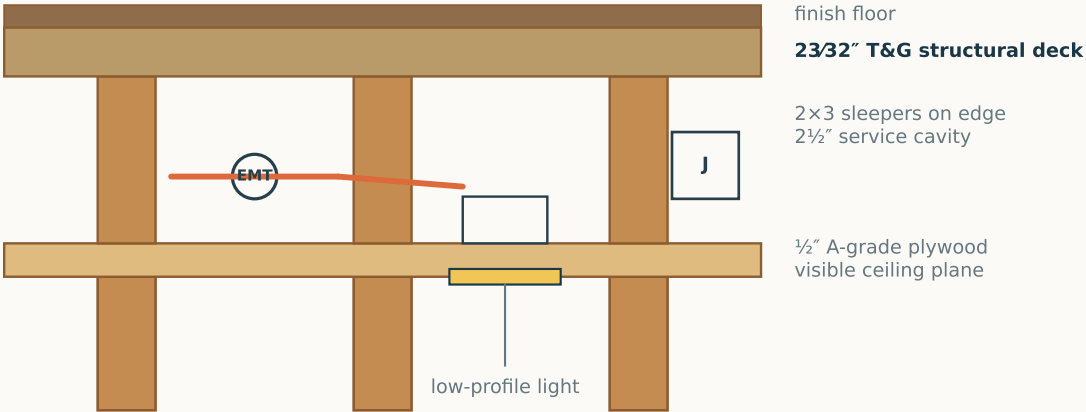
VISUAL RESULT: wood and steel share one crisp top plane; almost all hardware disappears in the W pocket.

Do not assume the cope provides bearing. Confirm end shear, seat, welds, web local strength, uplift and lateral restraint.

B HIDDEN SERVICE FLOOR — LIGHTS + WIRING ABOVE THE CEILING

Your layered idea works best when the upper panel is the structural deck and the lower panel is the visible ceiling.

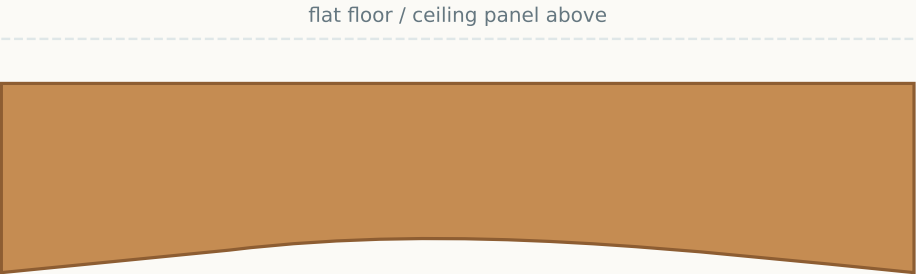
Dedicated conduit lanes cross over the W-beam top flange in the same cavity. Keep junction boxes accessible through a removable light or perimeter access panel.



If floor height must stay minimal: flat 2×4 sleepers give only 1 1/2". Use EMT / engineered steel protection; ordinary cable cannot sit safely in the screw zone.

C SHALLOW ARCH — MAKE THE SHAPE HONEST

The graceful version starts deeper; it does not carve capacity out of a 2×8 at midspan.



GOOD CONCEPT

engineered 2×10 / LVL / glulam profile; smooth CNC curve; minimum depth preserved at center



DON'T: field-cut the tension edge of the already-sized 2×8. The maximum cut lands near maximum bending.

At the 125 psf storage reading the current 2×8 study is already at DCR 0.96—there is effectively no reserve for an arch cut.

CONNECTION FAMILY — FROM QUIETEST TO MOST EXPRESSIVE

